

Emergence of Classical Mechanics from Discrete Causal-Information Networks: Ontological Mapping and Effective-Theory Limits within Status-Relational-Entropy (SRE) Dynamics

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Version: 1.0

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- Gemini Notebook (historical archive, no longer updated):
<https://notebooklm.google.com/notebook/ef52bf5a-f6d0-4a2a-aed4-b25d6520ab2c>
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According to the SRE principle, classical physical foundations originate from information statistics.

Reference baseline: SRE-v1.6 Axiom Suite (<https://doi.org/10.5281/zenodo.22077475>)

Note: This is a conceptual-framework paper focusing on establishing ontological correspondences rather than term-by-term rigorous analytical derivation of full textbook mechanics formulae. Three representative physical examples — inertia, force-mass-acceleration relation, and Hooke's law — illustrate emergent behaviour under coarse-graining, decoherence and topological-phase constraints. Supporting topological-evolution derivations are provided in the appendix. **Appendix derivations are illustrative topological sketches and shall not be treated as rigorous complete mathematical proofs; they serve only for physical intuition.**

Abstract

Classical mechanics is widely understood as a macroscopic effective limit of underlying microscopic physical descriptions. Within the Status-Relational-Entropy (SRE) dynamical framework, spacetime, mass and force are not primitive background entities but statistical emergent outcomes governed by dissipation-compensation duality within a discrete bidirectional causal-information network.

This paper establishes ontological mappings between core SRE network-level observables and central quantities of Newtonian classical mechanics. Three necessary conditions for the emergence of classical-like behaviour are identified: large-sample coarse-graining, sufficient environmental decoherence, and operating far from topological-phase-transition thresholds (the BBP spectral-rank phase-transition and causal-loop coherence saturation $\rho \rightarrow 1$). Three representative examples — inertia, the force-mass-acceleration relation, and Hooke's elastic law — demonstrate statistical emergence of classical-style behaviour when all conditions are satisfied, while also marking breakdown boundaries for each phenomenon.

Conservation of momentum and conservation of energy can be traced to global closed-loop symmetries of the underlying causal graph. This manuscript completes one conceptual segment of the SRE unified theoretical hierarchy: axiomatic causal-information network $\rightarrow$ composite-particle / gluon-sea mass generation $\rightarrow$ topological electrodynamics $\rightarrow$ emergent classical mechanics $\rightarrow$ cosmological BBP-RMT simulation constrained by spectroscopic observational datasets.

This work emphasises ontological correspondence instead of exhaustive analytical reproduction of every textbook formula in analytical mechanics. Illustrative topological-evolution derivations are supplied in the appendix for pedagogical intuition and are not regarded as rigorous core proofs of the paper.

Keywords: Status-Relational Entropy; causal-information network; emergent classical mechanics; coarse-graining; decoherence; effective theory; ontological mapping; Hooke's law; inertia

1 Introduction

Within conventional physics, Newtonian classical mechanics constitutes a highly successful macroscopic effective theory, while quantum mechanics describes microscopic degrees of freedom. A substantial body of existing research studies the emergence of classicality via decoherence and coarse-graining, explaining how familiar macroscopic laws arise from microscopic rules.

Status-Relational-Entropy (SRE) dynamics adopts a more fundamental starting point: there exists no pre-given continuous spacetime manifold as a primitive background. Space, time, mass and force are not ontologically fundamental material substances; they are all high-level render-layer emergent phenomena of discrete bidirectional causal-information networks subject to dissipation-compensation duality.

Prior SRE work has accomplished axiom construction, topological origin of composite particles together with gluon-sea-like mass amplification, graph-cohomology-based reconstruction of electrodynamics, and cosmological random-matrix simulation using SDSS/eBOSS spectroscopic datasets. Nevertheless, the conceptual bridge connecting discrete causal-network ontology to classical-mechanics phenomenology has not yet been systematically elaborated.

The present paper fills this conceptual gap. **This manuscript does not aim to derive every textbook formula of analytical mechanics term-by-term.** The main objectives are as follows:

1. State a set of necessary physical conditions under which classical-mechanics-like behaviour emerges from SRE causal-information networks;
2. Build an ontological-mapping table between network-level observables and physical quantities of classical mechanics;
3. Deploy three representative examples (inertia, $F \propto ma$, Hooke's law $F = -kx$) to demonstrate emergent behaviour together with corresponding breakdown thresholds;
4. Trace momentum-energy conservation to global closed-loop symmetries of causal graphs;
5. Delimit physical regimes where classical descriptions break down and one must fall back to the

underlying discrete-network description.

Illustrative pedagogical topological-evolution derivations are placed in Appendix A and are not treated as rigorous core proofs.

2 Brief Review of Core SRE Ontological Concepts

Only concepts essential for this manuscript are recapitulated; the full axiomatic system is given in the SRE-v1.6 Axiom Suite.

1. No primitive continuous spacetime background exists.
2. **Distance**: emergent macroscopic quantity representing topological-compensation overhead between causal nodes, originating from dissipation-compensation duality.
3. **Global evolution step ΔS** : discrete fundamental state-refresh cycle of the causal-information network; macroscopic time emerges as accumulated counting of these evolution steps.
4. **Mass**: originates from local closed causal-loops (strongly-connected subgraphs). Mass magnitude corresponds to total topological-path complexity of internal feedback loops within the local closed subgraph. Altering the configuration of such a subgraph demands network-compensation overhead, manifesting macroscopically as inertial impedance.
5. **Force**: not a primitive fundamental entity; it emerges statistically as a macroscopic topological-compensation gradient averaged over numerous microscopic causal-coupling perturbations.
6. Two characteristic classes of topological phase-transitions: the BBP spectral-rank phase-transition (cosmological dimensional crossover between 2D-holographic phase and 4D-unlocked-spacetime phase); and the causal-loop coherence-saturation transition $\rho \rightarrow 1$ during composite-particle formation.

3 Three Necessary Conditions for Emergence of Classical Mechanics

For macroscopic system-level behaviour to approximate Newtonian classical mechanics, **all three of the following conditions must hold simultaneously**. Violation of any single condition renders classical-level descriptions inappropriate and forces recourse to the underlying causal-network description.

1. Large-sample coarse-graining condition

The system contains a very large number of causal nodes and closed-loop subsystems. Random microscopic fluctuations of individual causal links are statistically averaged-out and suppressed at the macroscopic observable level. If the system possesses only few degrees-of-freedom, microscopic stochasticity cannot be averaged away and classical-like regular behaviour fails.

2. Effective decoherence condition

The system couples sufficiently strongly to the environmental causal-information network. Long-range quantum-phase coherence and large-scale superposition configurations are destroyed by environmental interactions. Persistent large-scale coherent entanglement breaks classical phenomenology.

3. Far-from-topological-transition-threshold condition

The system stays away from two classes of topological-transition points:

- Cosmological BBP spectral-rank phase-transition (2D-4D dimensional crossover);

- Causal-loop coherence-saturation $\rho \rightarrow 1$ associated with composite-particle formation.

Within transition regimes, compensation-operator topology undergoes abrupt structural reconstruction, and classical gradient-style force descriptions are no longer applicable.

When conditions 1 & 2 & 3 are jointly satisfied, statistical averaging over discrete causal-network dynamics yields behaviour corresponding to classical-mechanics phenomenology.

4 Ontological Mapping: SRE-Network Observables ↔ Classical-Mechanics Quantities

This table provides ontological correspondences, not strict point-wise numerical equality.

Quantity in classical mechanics	Ontological interpretation within SRE causal-information network
Inertial mass m	Total topological-feedback-path complexity inside a local closed causal-loop subgraph; network-compensation overhead required to alter its internal configuration, manifesting as inertial impedance.
Momentum p	Directed information-propagation flux of causal-loop clusters; global transport flow produced by synchronised evolution of large sets of network links.
Force F	Macroscopic topological-compensation gradient statistically averaged from numerous microscopic causal-coupling perturbations; an emergent gradient effect, not a primitive entity.
Spatial coordinate x	Coarse-grained average topological-geodesic causal-step count between different subgraphs; approximates continuous coordinate after coarse-graining.
Time t	Accumulated count of global discrete evolution steps ΔS ; approximates uniform continuous time within stable topological phases.
Kinetic energy E_k	Topological-flow overhead associated with directed propagation of causal-loop clusters.
Potential energy E_p	Latent compensation overhead stored in causal-coupling configurations among subgraphs.

4.1 Ontological Origin of Conservation Laws

- **Momentum conservation:** arises from global closed-loop symmetry of causal-network link interactions. Directed information-propagation flux cannot be locally created or annihilated out-of-nothing; flux can only be redistributed among subsystems. This yields macroscopic momentum conservation after coarse-graining.
- **Energy conservation:** originates from conservation of total topological-compensation overhead of the whole causal-information network. Transformations occur between internally-stored closed-loop configurations and open propagating-flow forms; total overhead magnitude is preserved.

5 Three Representative Physical Examples

Preamble: All examples below demonstrate behavioural correspondences under the triple emergence conditions. **They are not rigorous analytical mathematical proofs derived directly from axioms.** If any emergence condition breaks down, the described classical-style behaviour ceases to hold.

Topological-evolution formulae are given in Appendix A.

5.1 Example 1: Inertia

Classical phenomenon: A body maintains its existing state-of-motion in absence of external influences; larger inertial mass makes change-of-motion-state more difficult.

SRE ontological picture:

Inertia arises from total topological-feedback-path complexity inside local closed causal-loop subgraphs. A causal-loop cluster maintains its established internal link-synchronisation rhythm corresponding to macroscopic motion-state.

1. Without external perturbation: causal links preserve their existing synchronised configuration; no extra topological-compensation overhead is required. Macro-phenomenologically the object persists in its original motion-state — inertia.
2. To change motion-state: external coupling-perturbations must be injected to force re-synchronisation and rearrangement of large numbers of internal causal links, demanding substantial network-compensation overhead. Closed-loop subgraphs with richer internal feedback-paths (larger inertial mass) demand higher overhead for configuration-rearrangement; macroscopically they resist changes-of-motion more strongly.

Breakdown regimes:

If strong quantum-coherence dominates or the system enters the BBP spectral-rank topological transition, closed-loop topology itself mutates and classical inertial behaviour no longer applies.

5.2 Example 2: Force-mass-acceleration correspondence ($F \propto ma$)

Classical phenomenology: Larger applied force produces larger acceleration; for fixed force, larger inertial mass yields smaller acceleration, expressed as $a = F/m$.

SRE ontological picture:

Externally injected causal-coupling-perturbations produce macroscopic topological-compensation-gradients corresponding to the classical concept of force F .

1. Stronger external coupling-perturbations drive re-synchronisation of closed-loop cluster configurations within each global evolution step ΔS , giving larger effective macroscopic acceleration.
2. Inertial mass quantifies internal feedback-path complexity of the cluster. Higher complexity raises total compensation-overhead required for configuration-change. Under equal external perturbation strength, the rate-of-configuration-change is suppressed and effective acceleration becomes smaller.

Under full triple emergence conditions, statistically-averaged network-level behaviour yields the trend:

external-perturbation-strength $\propto$ inertial-mass $\times$ rate-of-configuration-change
which macroscopically corresponds to $F \propto ma$.

Important remark: This example illustrates behavioural matching rather than strict axiomatic proof. Breakdown occurs if coarse-graining, decoherence or far-from-transition conditions are violated.

5.3 Example 3: Hooke's elastic law $F = -kx$

Classical phenomenology: Within small-deformation elastic regime, restoring force is linearly proportional and opposite to displacement $F = -kx$. Beyond the elastic-limit linearity vanishes; plastic-deformation or fracture occurs.

SRE ontological picture:

An elastic solid consists of large numbers of mutually-coupled closed causal-loop clusters held in equilibrium by mediating causal links that establish balanced topological-geodesic configurations corresponding to macroscopic equilibrium-position.

1. **Deformation x :** External perturbation shifts local subgraph-clusters away from equilibrium topological-geodesic positions; average inter-subgraph causal-step separations are altered, mapping to macroscopic displacement.
2. **Origin of restoring force:** After displacement, mediating causal links generate topological-compensation-gradients. The network tends to pull subgraph-clusters back toward the original equilibrium configuration, producing restoring effects. For small offsets the compensation-gradient is approximately linear with displacement, yielding emergent Hooke-law behaviour $F = -kx$.
3. **Ontological interpretation of elastic limit:** When displacement grows too large, many mediating causal links break and re-wire; underlying network-topology reconstructs itself. The linear compensation-gradient relation is destroyed; macroscopically this corresponds to plastic-deformation or material fracture.

Breakdown regimes:

Large deformation causing link-rewiring; strong quantum coherence; entry into BBP topological-transition regime all invalidate classical elastic description.

6 Discussion

Within the SRE worldview, classical mechanics is not a fundamental set of axiomatic laws of nature. Instead it constitutes an effective-theory description that emerges only when the discrete causal-information-network satisfies three joint physical prerequisites: large-sample coarse-graining, sufficient environmental decoherence, and staying sufficiently far from major topological-phase-transition thresholds.

This paper does not claim to analytically derive every formula of Lagrangian-Hamiltonian analytical-mechanics term-by-term from SRE axioms. The contribution is ontological: establishing what each classical quantity corresponds to at causal-network level, stating emergence-conditions, demonstrating representative examples, and marking out explicit failure-regimes.

The conceptual hierarchy of the full SRE programme is conceptually closed:

- Discrete causal-information-network ontology
 - Composite-particle topology and gluon-sea-like mass-amplification
 - Graph-cohomology-based emergent electrodynamics
 - (coarse-graining + decoherence + far-from-transition) → emergent classical mechanics
 - Cosmological BBP-RMT spectral-rank-phase-transition simulation constrained by observational spectroscopic catalogues.

Important conceptual distinction: ontological mapping and behavioural-example matching is different from strict complete mathematical derivation. Future work can pursue more quantitative coarse-graining-operator formalisation to tighten the mathematical bridge between network-micro-dynamics and classical-level phenomenology.

7 Conclusions

1. Newtonian classical mechanics emerges from SRE discrete causal-information-network as an effective-theory description subject to three joint necessary conditions: large-sample coarse-graining, sufficient environmental decoherence, and operation far-removed from topological-phase-transition thresholds. Breaking any of these conditions invalidates classical-level description.
2. Core classical quantities (mass, momentum, force, space, time, kinetic / potential energy) can be given clear ontological interpretations at causal-network graph-level. Momentum- and energy-conservation laws trace back to global causal-graph closure symmetries.
3. Three representative examples — inertia, force-mass-acceleration trend, Hooke's law — illustrate how classical-style behaviours appear under the required emergence-conditions, and each example also defines its own breakdown threshold.
4. This work completes a conceptual link inside the SRE unified-framework; full term-by-term analytical derivation of all analytical-mechanics formulae is not performed here and remains a direction for follow-up research.

Appendix A Topological-Evolution Derivations (Illustrative Sketch, Not Rigorous Complete Proof)

Disclaimer: All topological-evolution formulae within this appendix serve only for physical-intuition pedagogy. **They shall not be treated as rigorous core proofs of this paper.** Complete strict formalisation is left for future work.

Notation Convention

1. ΔS : global discrete evolution step of the network;
2. $\mathcal{M}_{\text{topo}}$: local closed causal-loop subgraph, **topological-complexity (SRE proxy for inertial mass)**, counting total internal feedback-paths within the subgraph;
3. $\mathcal{G}_{\text{drive}}$: averaged external causal-coupling-perturbation strength (topological proxy for force);
4. $\mathcal{R}$: rate of internal-configuration rearrangement measured per global evolution step;
5. $\mathcal{D}_{\text{geo}}$: average topological-geodesic step-count between subgraphs (topological proxy for displacement);
6. $\mathcal{K}_{\text{topo}}$: topological-stiffness coefficient for the set of mediating causal links;
7. $\langle \cdot \rangle$: large-sample coarse-graining statistical-average operator;
8. All computations inside this appendix assume the three emergence-conditions hold: large-sample coarse-graining, decoherence, far from topological phase-transitions.

A1 Topological formulation of inertia

Closed-loop subgraph maintains its original configuration with zero external perturbation $\mathcal{G}_{\text{drive}} = 0$:

$$\left\langle \frac{\partial \mathcal{M}_{\text{topo}}}{\partial \Delta S} \right\rangle = 0$$

Under external driving, topological-compensation overhead consumed by configuration-rearrangement scales with topological complexity:

$$\mathcal{C}ost_{\text{comp}} \propto \mathcal{M}_{\text{topo}} \cdot \mathcal{R}$$

Larger $\mathcal{M}_{\text{topo}}$ demands higher topological-compensation overhead for configuration change, corresponding to larger inertia.

A2 Topological-evolution relation for force-mass-acceleration ($F \propto ma$)

Under statistical averaging, external causal-coupling perturbation yields a topological driving term:

$$\langle \mathcal{G}_{\text{drive}} \rangle \propto \mathcal{M}_{\text{topo}} \cdot \mathcal{R}$$

$\mathcal{R}$ denotes configuration-rearrangement rate per evolution-step, the topological proxy for acceleration. Mapping topological proxies onto classical render-layer observables:

$$F \longleftrightarrow \langle \mathcal{G}_{\text{drive}} \rangle, \quad m \longleftrightarrow \mathcal{M}_{\text{topo}}, \quad a \longleftrightarrow \mathcal{R}$$

This yields the emergent trend relation:

$$\mathbf{F} \propto m\mathbf{a}$$

Remark: This expresses proportional-trend correspondence, not strict equality with known constant of proportionality; proportional constants originate from global network topological rigidity and require global spectral inputs.

A3 Topological-evolution derivation for Hooke's law

$\mathcal{D}_{\text{geo},0}$: topological-geodesic step-count between subgraphs at equilibrium;

$\Delta \mathcal{D}_{\text{geo}}$: topological-geodesic offset (topological proxy for displacement):

$$\Delta \mathcal{D}_{\text{geo}} = \mathcal{D}_{\text{geo}} - \mathcal{D}_{\text{geo},0}$$

Topological restoring-compensation gradient generated by mediating causal links:

$$\langle \mathcal{G}_{\text{restore}} \rangle = -\mathcal{K}_{\text{topo}} \cdot \Delta \mathcal{D}_{\text{geo}}$$

Mapping onto classical observables:

$$F \longleftrightarrow \langle \mathcal{G}_{\text{restore}} \rangle, \quad k \longleftrightarrow \mathcal{K}_{\text{topo}}, \quad x \longleftrightarrow \Delta \mathcal{D}_{\text{geo}}$$

For small offsets without link breaking-rewiring, we obtain the emergent relation:

$$\mathbf{F} = -k\mathbf{x}$$

Breakdown topological condition: When $|\Delta\mathcal{D}_{\text{geo}}|$ exceeds the tolerance threshold of mediating links, mediating causal links break and re-wire; $\mathcal{K}_{\text{topo}}$ is no longer constant and the Hooke-law topological relation collapses.

A4 Topological form of conservation laws

Total topological-compensation overhead $\mathcal{U}_{\text{total}}$ of the global causal-information network is conserved:

$$\frac{d\langle\mathcal{U}_{\text{total}}\rangle}{d\Delta S} = 0$$

Information-propagation flux (topological proxy for momentum) satisfies global closed-loop constraints: only redistribution among subsystems occurs, total flux remains unchanged.

References

1. SRE Dynamics Axiom Suite v1.6, Zenodo archive.
2. Zurek, W.H. Decoherence and the transition from quantum to classical.
3. Hoel, E.P. Causal emergence and coarse-graining in complex networks.
4. Ehrenfest theorem: quantum-classical correspondence principle.
5. SRE-v6.2-rev: cosmological BBP-RMT spectral-rank-phase-transition simulation with SDSS/eBOSS spectroscopic datasets.